MILITARY SPECIFICATION

RING BUOY, LIFESAVING, AUTO-INFLATABLE

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 scope. This specification covers the requirements for a water-activated, self-inflating ring buoy enclosed in a shell that is suitable for throwing from a ship or aircraft to a victim in the sea.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY
MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging.
MIL-C-601 - Cylinders, Carbon Dioxide Filled, Technical.
MIL-L-3454 - Life Preservers, Packaging of.
MIL-C-83489 - Cloth, Coated, Nylon, Polyurethane Coated.

STANDARDS

FEDERAL
FED-STD-595 - Colors.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4220

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MIL-STD-l05 – Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II- Internally Excited).

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the document which is DoD adopted shall be that listed in the issue of the DoDISS specified in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 117 - Standard Method of Salt Spray (Fog) Testing. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials.

3.2.1 Coated cloth. The coated cloth used in the construction of the ring buoy shall conform to type I of MIL-C-83489, except that the tear strengths shall be 3.75 pounds warp, 2.75 pounds fill, and 45 pounds adhesion.

3.2.2 Automatic inflation system. Each ring buoy shall be provided with a water-activated carbon dioxide (C02) inflation device that will perform as specified in 4.6.1.

3.2.3 C02 cylinders. The cylinders used in the inflation system shall perform as specified in 4.6.2.

3.2.4 Reflective tape. Scotchlite Brand Solas Grade number 3150 reflective adhesive tape with number 2141 primer, Minnesota Mining and Manufacturing Company, or its equivalent shall be used. The tape shall be approximately 2 inches wide and wrapped completely around the ring buoy in four equally spaced places (see figure 1).

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3.3 Design.

3.3.1 Ring buoy. The ring buoy shall be ring or horseshoe shaped with an inside diameter of 15 + 1/2 inches and an outside diameter of 25 + 1/2 inches. If the buoy is horseshoe shaped it shall have a strap system for closing the open end of the buoy.

3.3.2 Plastic shell. In its uninflated state, the ring buoy shall be encased in a durable plastic shell that shall withstand the drop test specified in 4.6.4. The plastic shell shall have a maximum length of 12 inches and a maximum diameter of 4.5 inches (see figure 2). The plastic shell shall be attached to the buoy by a cord 16 inches long to act as sea anchor.

3.4 Construction. Air retaining seams in the ring buoy and attachment of all other parts to the ring buoy shall be constructed by heat sealing.

3.4.1 Sealed areas. The construction of heat sealed seams and the attachment of all parts shall utilize techniques and precautions to develop optimum properties of sealed areas as specified in MIL-C-83489.

3.5 Rsistance to damage. There shall be no damage experienced by either the outer shell or the ring buoy when subjected to a drop test, as specified in 4.6.4.

3.6 Vibration. The encapsulated ring buoy shall be vibrated as specified in 4.6.5.

3.7 Buoyancy. The ring buoy shall have a minimum buoyancy of 20 pounds when tested as specified in 4.6.6.

3.8 Weight. Total weight of the ring buoy encapsulated in the plastic shell shall not exceed 2.5 pounds.

3.9 Shelf life. The packaged, encapsulated ring buoy shall have a minimum shelf life of 5 years.

3.10 Color. Unless otherwise specified (see 6.2.1), the coated cloth and plastic shell color shall be fluorescent orange, in accordance with FED-STD-595, color chip number 38903.

3.11 Marking. Identification markings shall be made by the silk screen method in letters and numerals 1/4 inch high. The ink shall be a contrasting color, waterproof, and cause no deterioration of fabric.

3.11.1 Identification. The identification shall consist of the name of the item, serial number, the name or trademark of the manufacturer, date of manufacture (month and year) and the contract number. The identification shall be located inside the protective package and on the ring buoy.

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3.11.2 Operating instructions. Operating instructions shall be supplied by the contractor on a paper inserted into the transparent shipping package in such a manner that the instructions are visible at all times.

3.12 Workmanship. Workmanship shall be in accordance with standard manufacturing practice covering this class of work.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified here in, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection. First article inspection shall be performed on samples submitted as specified in 4.3.1. First article inspection shall consist of the examinations of 4.6.1 through 4.6.9.

4.3.1 Samples for first article inspection. The following samples shall be submitted for first article inspection as required.

(a) Four complete, packaged ring buoys encased in their shells.
(b) Two cured pieces of ring coating compound, 6 inches long by 6 inches wide and 0.07 to 0.08 inch thick.
(c) Two yards of each of all coated and uncoated cloth, used in the ring buoy. The name of the supplier and his code number shall be furnished along with the samples for this item.
(d) Four seam adhesion samples of the ring cloth, 12 inches wide by 12 inches long, with a 1/8 inch wide seam running the entire width of the sample along the centerline.
(e) Two heat sealed samples, 7 inches wide by 8 inches long, prepared coated face to coated face.

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(f) Four sets of all hardware, including inflation system, used in the complete ring buoy.

4.3.2 First article inspection report and inspection procedure. Unless otherwise specified (see 6.2.1), the contractor shall prepare a first article inspection report and a first article inspection procedure in accordance with the data ordering document specified (see 6.2.2).

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.2, 4.4.3, 4.4.5.l and 4.4.5.2.

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

4.4.2 Leakage test samples. Ring buoys shall be 100 percent tested for leakage in accordance with 4.6.3 prior to packing into the plastic shell. Ring buoys not conforming to 4.6.3 may be reworked prior to retesting, but no air retaining seam shall be reworked more than once.

4.4.3 Adhesion test samples. At the beginning of each day's production, one buoy from each 100 production buoys sealed shall be tested in accordance with the applicable tests of 4.6.9. If, during the cycle, buoys of fabric from a different fabric coater are to be sealed, a fresh sample shall be tested and the cycle count returned to zero. If failures occur during any sample tests, production shall be halted and corrective action shall be taken to determine and correct the failure.

4.4.4 Inspection lot. A lot shall consist of all ring buoys submitted for inspection at one time.

4.4.5 Examination of the end item. A random sample of ring buoys shall be selected from each lot in accordance with inspection level 11 of MIL-STD-105 for the examination specified in 4.4.5.2. The acceptable quality level (AQL) shall be 1.0 for all defects.

4.4.5.1 End item test. A random sample of ring buoys shall be selected from each lot in accordance with inspection level II of MIL-STD-105 with an AQL of 10. Each sample selected shall be subjected to the specified tests and performed in the order shown as follows:

Test	Test paragraph
Automatic inflation	4.6.1
Drop test	4.6.4
Buoyancy	4.6.6

4.4.5.2 Visual examination. The finished ring buoys shall be examined for the defects classified in accordance with table I. The AQLs and inspection levels of 4.4.5 apply for critical, major and minor defects.

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TABLE I. Classification of defeats.

Classification	Defects
Critical:	
1.	Ring buoy material defective; evidence of splits, tears, cuts, holes, patched holes, fraying, burns, or not as specified.
2.	Ring buoy seams defective, incomplete, or not as specified.
3.	Inflation system not secure; securing patches not as required.
4.	Inflation assembly not securely attached.
5.	Parts imparing the function of the ring buoy as an item of lifesaving equipment.
Major:	
101	Hardware missing, or not secure; evidence of cracks, pits, or sharp surfaces that may cut bladder or are incapable of functioning.
102	Packaging and packing not as specified.
103	Evidence of reworking air retaining seams more than the allowable number of times.
104	Operation instructions missing, illegible, or not located as specified.
Minor:	
201	Marking, manufacturer's name, date of manufacture and control number missing, incorrect, illegible, not permanent, or not located as specified.
202	Completed ring buoy not clean; one or more spots or stains which are clearly noticeable and not removable.
203	Seams twisted, puckered, or pleated.
204	Reflective tape improperly applied or not as specified.

4.5 Atmospheric conditions. Unless otherwise specified in the contract or order (see 6.2.), all inspections and tests be conducted at prevailing atmospheric pressure, or less. Celsius (°C) and a relative humidity of 80 percent

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4.6 Methods of inspection.

4.6.1 Automatic inflation operation. The automatic inflation operation of the ring buoy shall be tested by throwing the ring buoy encased in its shell into a body of water. The ring buoy shall fully inflate within 10 seconds after hitting the water. Failure of the plastic shell to separate, evidence of leaks, partial inflation or no inflation after 10 seconds shall constitute noncompliance

4.6.2 CO_2 cylinders. The CO_2 filled cylinders shall conform to the following quality conformance inspection tests of MIL-C-601:

(a) Elevated temperature.
(b) Leakage.
(c) Hydrostatic pressure.
(d) Rupture.
(e) Puncturing.

4.6.3 Leakage. The ring buoy shall be fully inflated with air through the inflation system to a pressure of 2 pounds per square inch (lb/in^2) and allowed to stand for a minimum of 4 hours. At the end of this time the pressure shall be measured and corrected for any change in temperature or barometric pressure. The pressure drop shall be not greater than 0.25 lb/in^2. Seams and sealed areas shall remain intact and shall show no signs of separation.

4.6.4 Drop test. The encapsulated ring buoy shall be dropped three times from a height of 6 feet onto a fixed steel plate. Failure of the ring buoy to function as specified in 4.6.1 after this test shall constitute noncompliance

4.6.5 Vibration test. The encapsulated ring buoy shall be subjected to a type 1 vibration test of MIL-STD-167-l. Upon successful completion of this test, the ring buoy shall be subjected to the auto-inflation test of 4.6.1.

4.6.6 Buoyancy. Immediately upon completion of the auto-inflation test (see 4.6.1), the ring buoy shall be tested as follows to determine the buoyancy. In this test, the inflated ring buoy shall be placed in a cage of adequate size and of sufficient weight to submerge the ring buoy. This assembly shall be suspended from a direct reading scale, in fresh water, at room temperature, with the top of the cage 2 inches beneath the surface of the water. The scale reading shall be taken as soon as practicable and the buoyancy calculated as the apparent weight of the cage and inflated ring buoy subtracted from the previously determined underwater weight of the assembly without the inflated ring buoy. This calculated buoyancy shall be a minimum of 20 pounds.

4.6.7 Temperature. The ring buoy shall be placed in a circulating air oven maintained at a temperature of 65 $\pm$ 3°C for a minimum of 20 hours and then placed in a cold chamber at a temperature of minus 30 + 3°C for a minimum of 20 hours. Upon completion of this test the ring buoy shall pass the test specified in 4.6.1.

4.6.8 Salt spray. The encapsulated device in its plastic bag shall be subjected to a 100-hour salt spray test in accordance with ASTM B 117. Immediately following the salt spray test, the device shall be subjected to the tests specified in 4.6.1 and 4.6.3.

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4.6.9 Adhesion. The sample heat sealed bladder seams shall be subjected to the adhesion test specified in MIL-C-83489.

4.7 Inspection of Packaging. Sample packages and packs, and the inspection of the packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5* PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation. Ring buoys shall be packaged level A or C as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each ring buoy shall be packed in a close fitting, transparent , heat-sealed bag conforming to MIL-B-117, type I, class E, style 2. Identification marking and operating instructions shall be placed in the bag prior to heat sealing.

5.1.1.2 Intermediate packaging. Six units of protected ring buoys shall be packed in a close fitting fiberboard box conforming to PPP-B-636, class weather resistant, variety SW, grade V or W with type optional. Full height fiberboard divider of the same material as that of the box shall be placed between the ring buoys. Box closure shall conform to method V of the appendix to PPP-B-636.

5.1.2 Level C.

5.1.2.1 Unit packaging. Unit packaging shall be as specified in 5.1.1.1.

5.1.2.2 Intermediate packaging. Intermediate packaging shall be as specified in 5.1. 1.2, except that the boxes shall be of the non-weather resistant, domestic class, variety SW, minimum grade 200 with method I closure applicable.

5.2 Packing and marking. Ring buoys packaged as specified in 5.1 shall be packed level A, B or C as specified (see 6.2.1) and marked in accordance with MIL-L-3454. Marking shall include the bar code requirement of MIL-STD-129.

6. NOTES

6.1 Intended use. The mechanically operated auto-inflatable ring buoy covered by this specification is designed to be thrown, in an emergency situation, to a victim in the sea. It is intended to be used as an immediate auxiliary to other lifesaving equipment.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color, if other than fluorescent orange (see 3.10).
(c) If first article inspection report and procedure is to be submitted (see 4.3.2).
(d) If atmospheric pressure, temperature or relative humidity during tests is other than specified (see 4.5).
(e) Level of packaging and packing required (see 5.1 and 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.3.2	First article inspection report	DI-T-4902	
4.3.2	First article inspection procedure	DI-T-4901	----

(Data item descriptions related to this specification, and Identified in section 6 will be approved and listed as such in DoD 5000.19L. , Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the items should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves

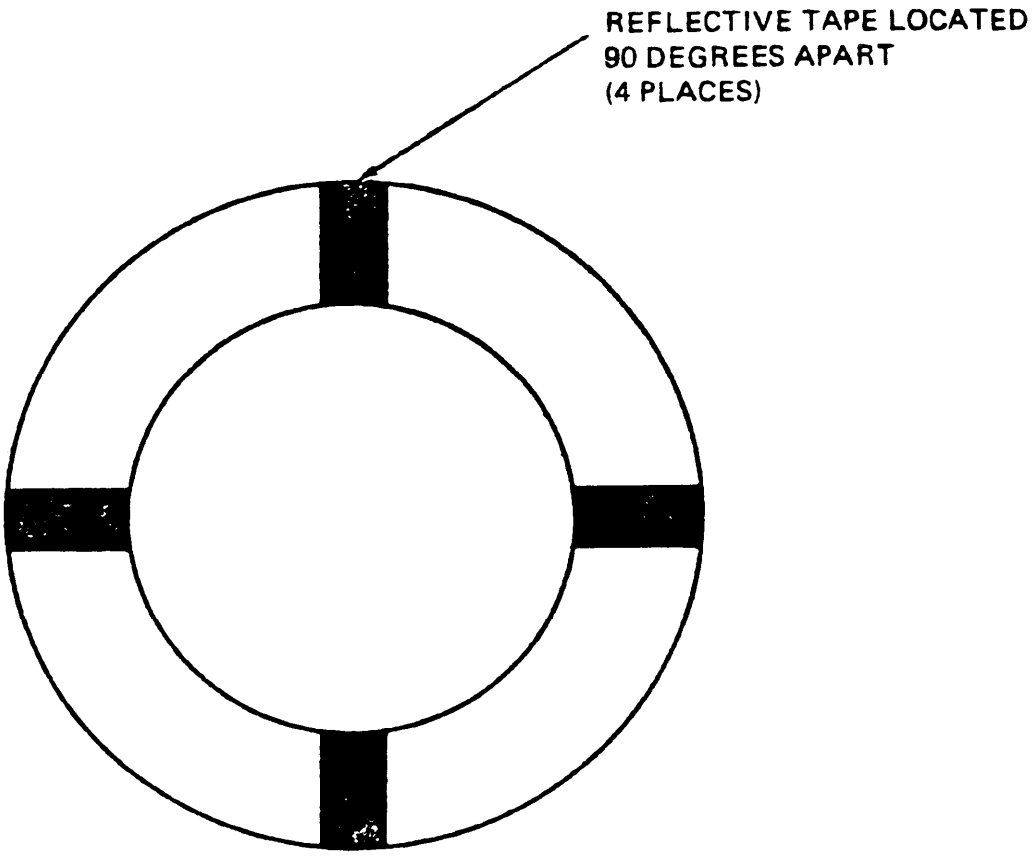
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the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
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FIGURE 1. Reflective tape location on ring buoy.

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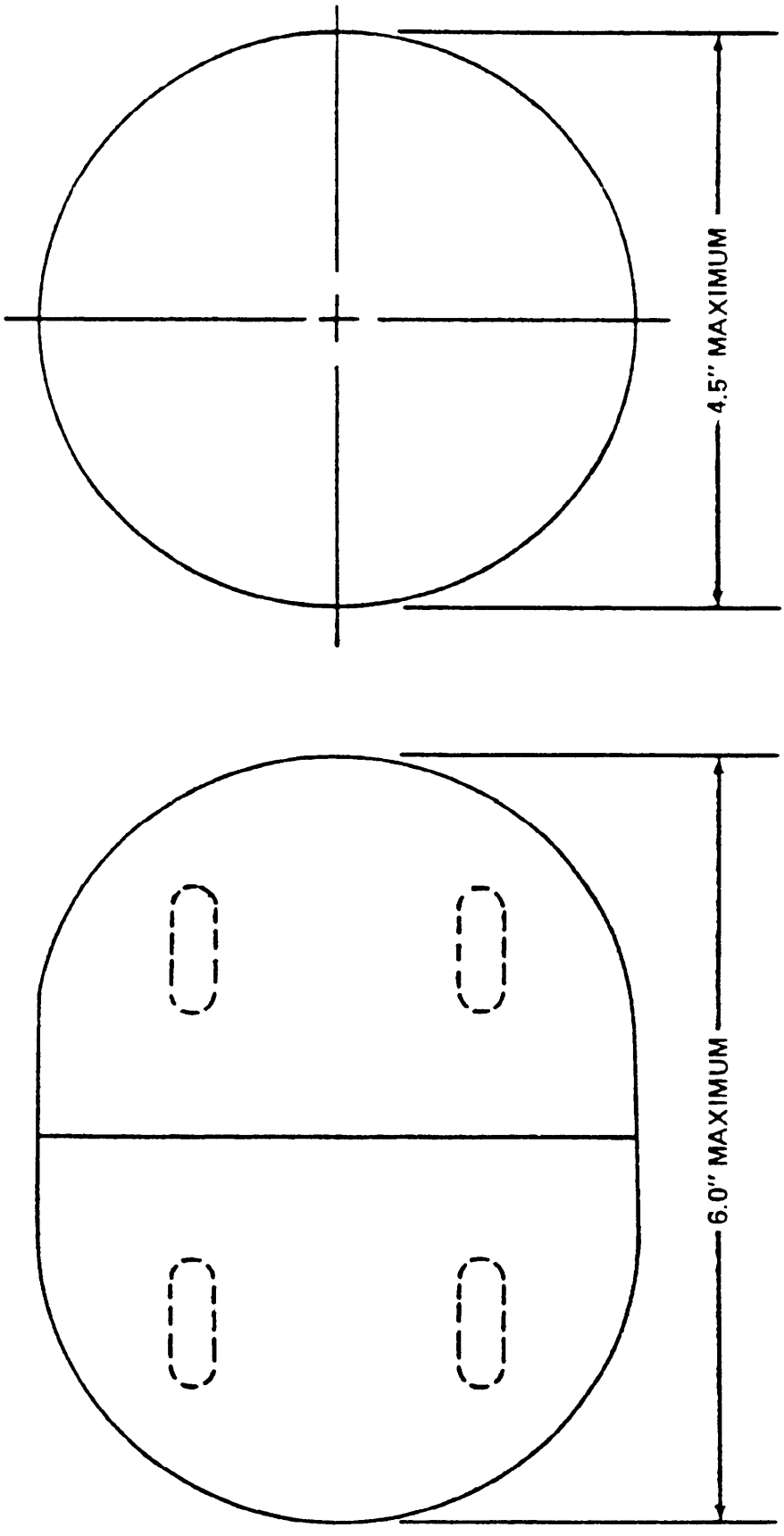


FIGURE 2. Maximum dimensions of enclosed ring bouy.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-24661(SH)	RING BUOY, LIFESAVING, AUTO-INFLATABLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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